

Advanced (Habanero)

Q1. What is the value of the digit 7 in 74,317?

- (A) 7
- (B) 70
- (C) 700
- (D) 7,000
- (E) 70,000

Q2. Round 45,678 to the nearest thousand.

- (A) 45,000
- (B) 46,000
- (C) 40,000
- (D) 50,000
- (E) 44,000

Q3. Write 943,210 in expanded form.

- (A) $900,000 + 40,000 + 3,000 + 200 + 10$
- (B) $900,000 + 40,000 + 3,000 + 200 + 1$
- (C) $900,000 + 40,000 + 3,000 + 2,000 + 10$
- (D) $900,000 + 40,000 + 3,000 + 200 + 0$
- (E) $900,000 + 40,000 + 4,000 + 200 + 10$

Q4. What is the standard form of $5 \cdot 10^4 + 2 \cdot 10^3 + 7 \cdot 10^0$?

- (A) 52,007
- (B) 50,027
- (C) 52,700
- (D) 52,000
- (E) 50,700

Q5. Round 0.4567 to the nearest hundredth.

- (A) 0.46
- (B) 0.45
- (C) 0.47
- (D) 0.50
- (E) 0.48

Q6. Write 0.053 in standard form.

- (A) $53 \cdot 10^{-3}$
- (B) $5.3 \cdot 10^{-2}$
- (C) $530 \cdot 10^{-4}$
- (D) $0.053 \cdot 10^1$
- (E) $53 \cdot 10^{-2}$

Q7. What is the place value of 8 in 85,039?

- (A) Ones
- (B) Tens
- (C) Hundreds
- (D) Thousands
- (E) Ten thousands

Q8. Round 987,654 to the nearest hundred thousand.

- (A) 1,000,000
- (B) 900,000
- (C) 1,000,000
- (D) 100,000
- (E) 900,000

Open Ended Questions (FRQ)

Q9. Write 6,542,819 in expanded form and explain the reasoning behind it.

Q10. Round 4,239.54 to the nearest tenth. Justify your answer with a numerical explanation.

Chef's Tips

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- Tip 1: Students should be familiar with the concept of place value and how it applies to multi-digit numbers.
- Tip 2: Pay close attention to the number of significant figures in each question.
- Tip 3: Make sure to use correct mathematical notation and provide clear explanations for FRQs.